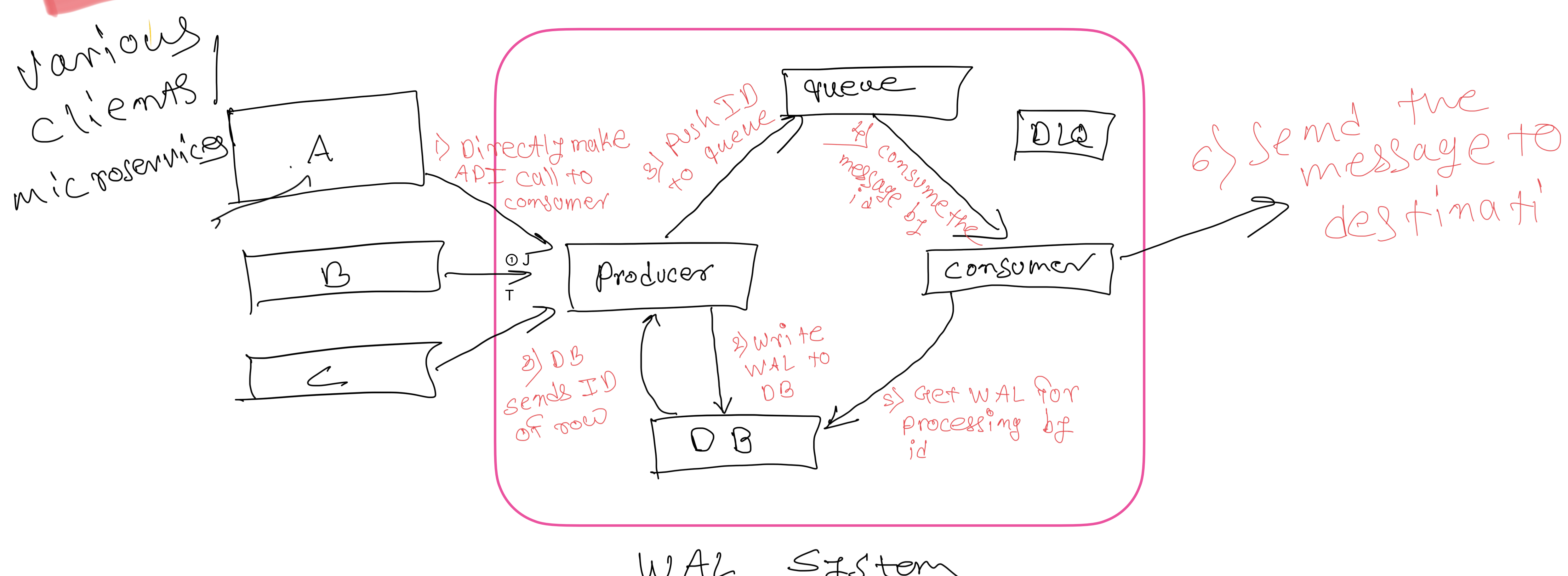


Two types to process

1) WAL First

2) WAL on Failure → To ensure retry & durability

WAL First

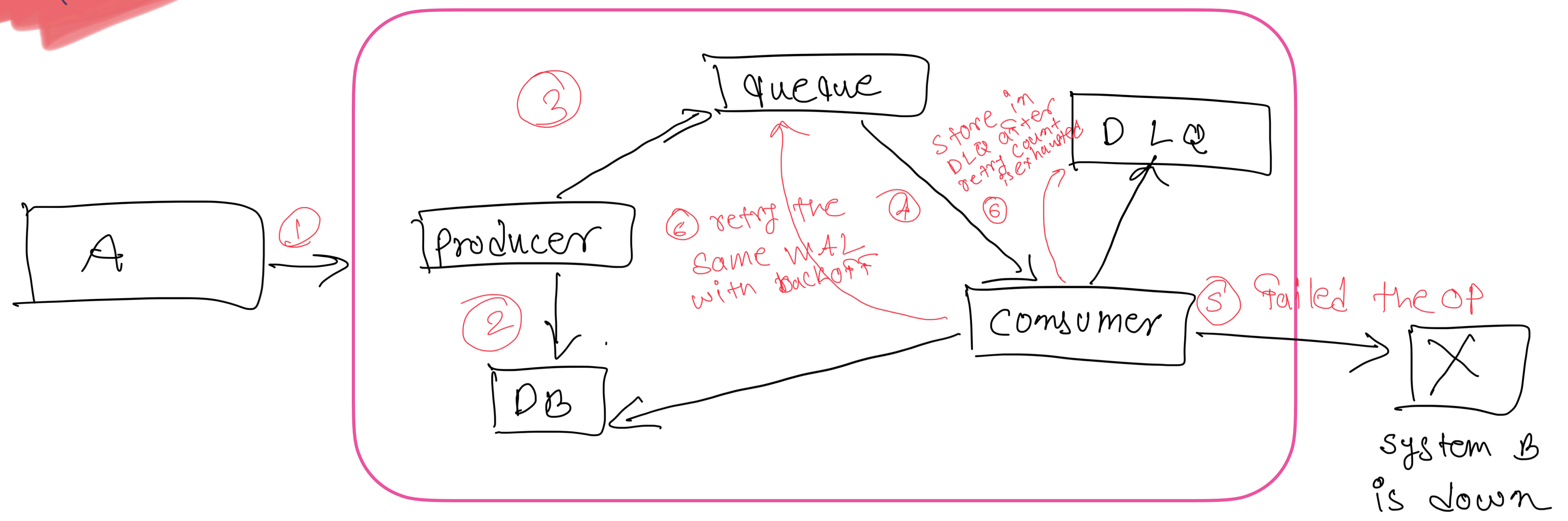


★ WAL is stored in DB when doing Batch processing

- 1) Firstly write All WALs to DB
- 2) Send confirmation to consumer
- 3) consumer gets the WAL and execute it
- 4) Delete the WALs from DB

★ Else we store WAL in queue for single processing to avoid decoupling

DLQ → Dead Letter Queue



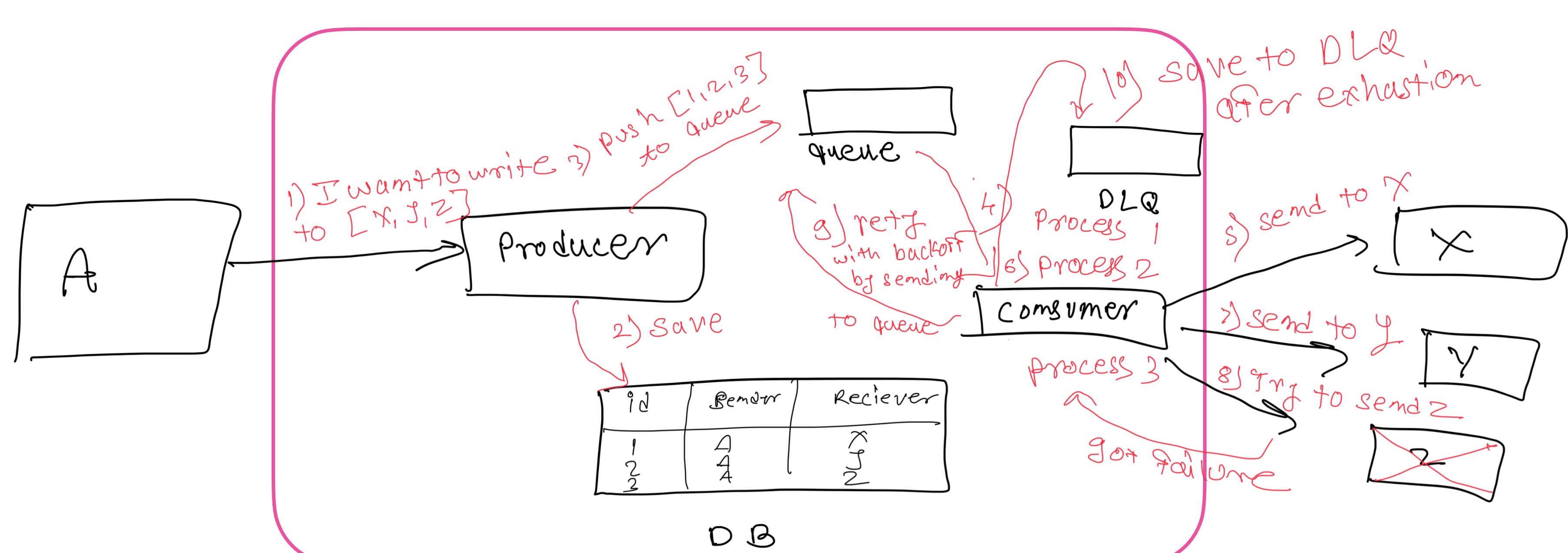
★ So when an endpoint where consumer wants to write is down

1) on error forward the WAL to the separate delay queue or same queue with some delay

2) IF we exhaust the amount of retries allowed then write the WAL to DLQ.

3) DLQ should be handled manually

Let's understand with example



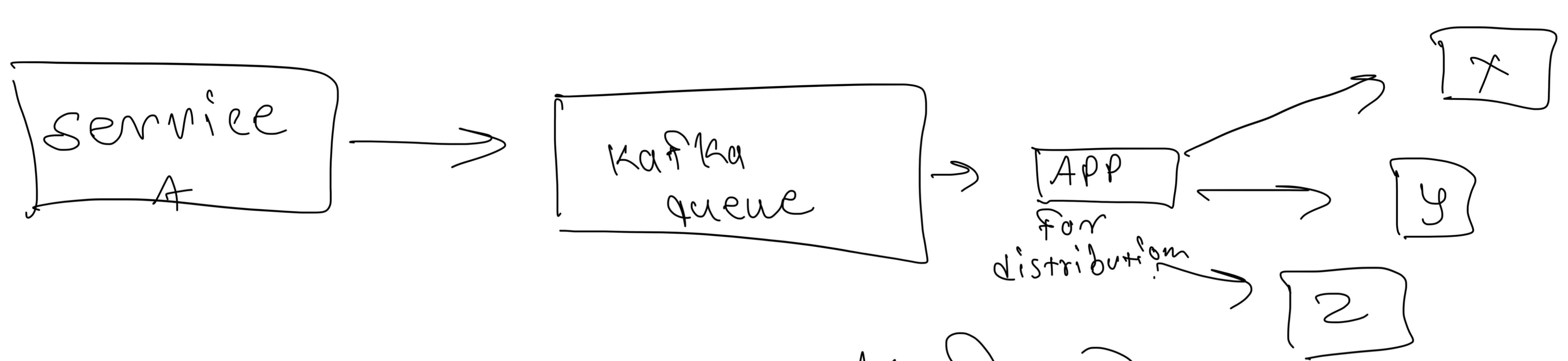
★ When service 'z' is down we'll retry certain times & after that we'll write to the DLQ.

* IF we get something in DLQ then it's a possible outage so inform the on-call eng.

* IF write to DLQ fails then we have a outage need to inform respective team.

2) WAL on Failure

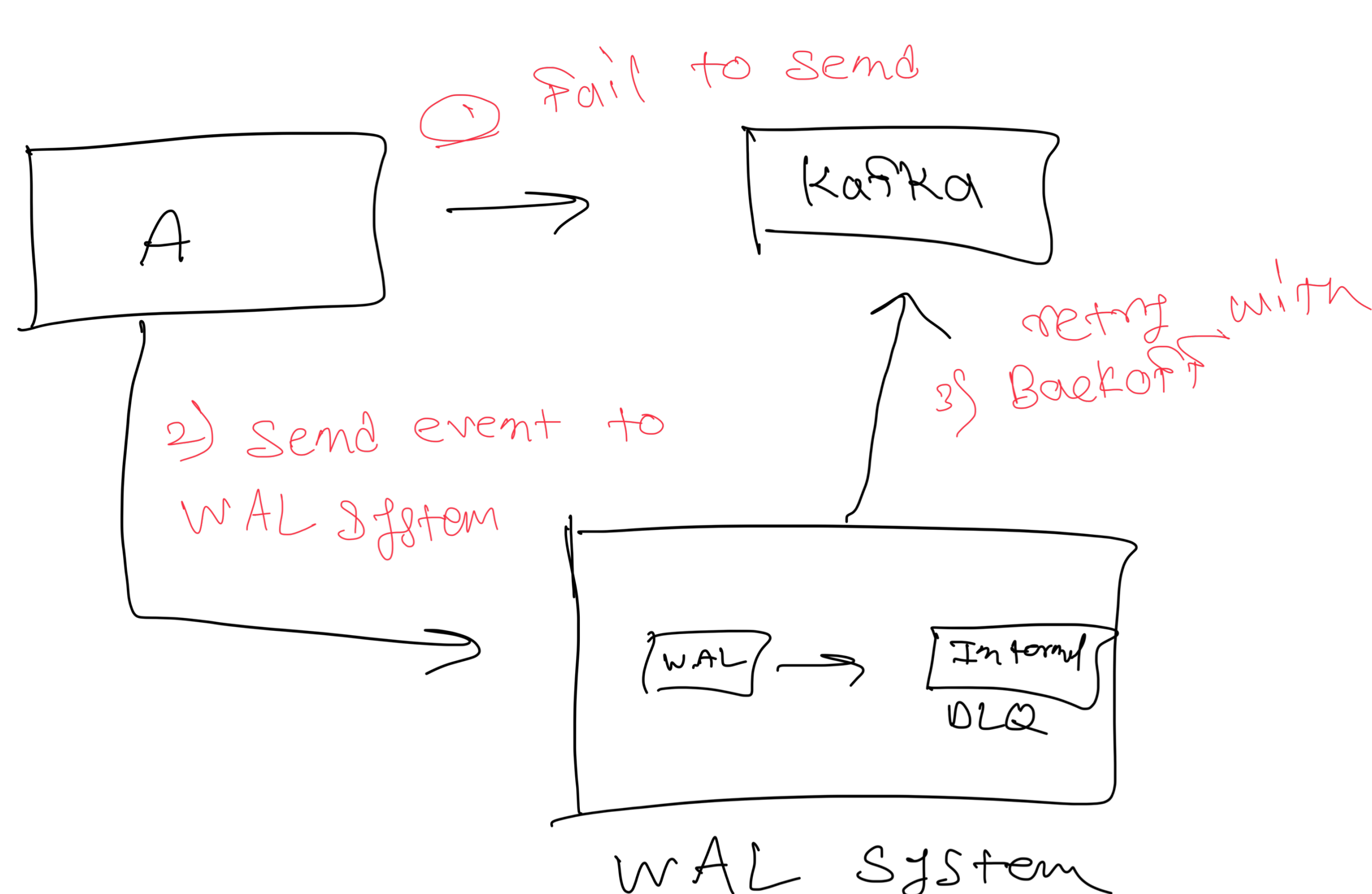
In this mode we will use WAL for failure handling



Ideal work flow

★ Failure modes

① Client is failing to send event to Kafka



② Failure when Kafka can't reach destination

